SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT TEMPLATE

1. Student Details

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Division	A
Batch	A1
Roll No.	202501110025
Date of Submission	23-10-2026

2. Practical Details

Practical No.	10
Practical Title	IMPLEMENT CLASS TEMPLATE AND FUNCTION TEMPLATE
Date Performed	23-10-2026
Faculty Name	Dr. Khushal Khairnar

3. Problem Statement

Write a C++ program to demonstrate a class template and a function template. Create a Calculator class template that stores two values and performs addition and multiplication. Also create a function template that returns the maximum of two values.

4. Objectives

1. Understand the concept of templates in C++.
2. Implement a class template for operations on two values.
3. Implement a function template to find the maximum of two values.
4. Demonstrate template use with integer and floating-point values.

5. Expected Learning Outcomes

After completing this practical, the student will be able to:

1. Explain generic programming and templates.
2. Define and instantiate a class template.
3. Define and call a function template with different data types.
4. Reuse code for multiple data types without duplicating logic.

6. Theory / Concept

Template: A C++ feature for writing generic code that works with different data types.

Class Template: A class defined with a template parameter, allowing its data members and functions to work with different types.

Function Template: A function pattern that can operate on values of different types.

Generic Programming: A programming approach that emphasizes reusable algorithms and data structures independent of a specific type.

7. Algorithm / Pseudocode

START

1. Define a class template Calculator with type parameter T.
2. Store two values of type T in the class.
3. Define member functions to calculate the sum and product and display the values.
4. Define a function template findMaximum(T x, T y).
5. Create an integer Calculator object and display its operations.
6. Call findMaximum with integer values and floating-point values.
7. Display all results.

STOP

8. Technology / Tools / Libraries Used

Programming Language	C++
IDE / Platform	Dev C++
Libraries / Packages	iostream

9. Implementation / GitHub Repository

GitHub Repository URL	https://github.com/pranav-4797/OOP-Practical-Assignment
Repository Name	OOP-Practical-Assignment
Practical Folder	Practical 10
Branch Used	main
Commit ID / Latest Commit	Not provided

Repository Contents Used

The source file is provided separately as “Practical_10_Templates.cpp”. Add it to the Practical 10 folder of the repository if required.

10. Test Cases / Input and Output

Test Case	Operation	Expected Output	Status
1	Calculator<int>(10, 5)	Sum = 15; Product = 50	Pass (sample)
2	findMaximum(12, 20)	20	Pass (sample)
3	findMaximum(4.5, 2.8)	4.5	Pass (sample)

11. Results / Observations

The program demonstrates both class and function templates. The Calculator class template performs addition and multiplication for integer values, while the findMaximum function template works with both integer and floating-point values.

12. Source Code

```
#include <iostream>
using namespace std;

template <class T>
class Calculator {
private:
T a, b;

public:
Calculator(T x, T y) {
a = x;
b = y;
}

T add() {
return a + b;
}

T multiply() {
return a * b;
}
```

```
void display() {
cout << "First value: " << a << endl;
cout << "Second value: " << b << endl;
cout << "Sum: " << add() << endl;
cout << "Product: " << multiply() << endl;
}
};

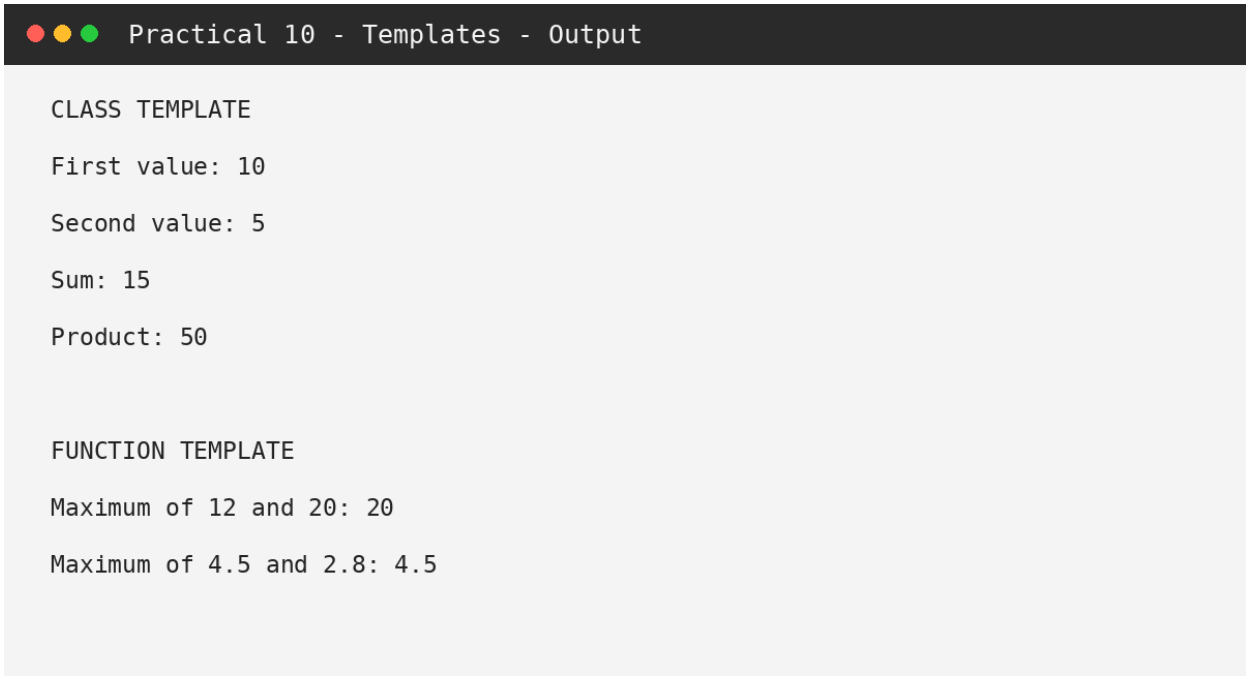
template <typename T>
T findMaximum(T x, T y) {
return (x > y) ? x : y;
}

int main() {
cout << "CLASS TEMPLATE" << endl;
Calculator<int> intCalc(10, 5);
intCalc.display();

cout << "\nFUNCTION TEMPLATE" << endl;
cout << "Maximum of 12 and 20: " << findMaximum(12, 20) << endl;
cout << "Maximum of 4.5 and 2.8: " << findMaximum(4.5, 2.8) << endl;

return 0;
}
```

Screenshots / Output



This is a representative terminal-style output image. Run the source code in Dev-C++ to capture an actual IDE screenshot if required.

13. Conclusion

This practical successfully demonstrates a class template and a function template in C++. Templates make code reusable for different data types. The class template performs arithmetic operations, and the function template finds the maximum value for integer and floating-point arguments.

14. Assignment Questions

Q1. What is a template in C++?
Answer: A template allows generic code to be written for multiple data types.

Q2. What is a class template?
Answer: A class template defines a class using type parameters.

Q3. What is a function template?
Answer: A function template defines a generic function that can work with different data types.

Q4. What is the advantage of templates?
Answer: They improve code reuse and reduce repeated implementations for different types.

Q5. Which keyword is used to declare a template?
Answer: The template keyword.

15. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

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Signature	